Pursuer Threat Area Prediction Based on Heteroscedastic Sparse Variational Gaussian Processes

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Abstract: In pursuer-evader scenarios for flight vehicles, predicting the pursuer's threat area can substantially improve the evader's strategic planning efficiency. However, flight environments often exhibit highly complex noise characteristics, and inadequate noise modeling can severely degrade prediction performance. To address this challenge, we propose a threat area prediction algorithm based on heteroscedastic sparse variational Gaussian processes (HSVGP). By employing a radial basis function network to construct a heteroscedastic likelihood function, the proposed method captures input-dependent noise characteristics, thereby substantially enhancing prediction accuracy in complex environments. Furthermore, by leveraging HSVGP's intrinsic uncertainty quantification capabilities, the algorithm provides principled estimates of threat area boundaries. Simulation results demonstrate that the HSVGP significantly outperforms its homoscedastic counterpart under complex noise conditions.

Key Words: threat area prediction, pursuit-evasion game, evasion, Gaussian process

1 Introduction

The pursuit-evasion game in flight vehicle scenarios represents a typical non-cooperative adversarial problem. Accurate prediction of the pursuer's maneuvering behavior can provide effective guidance for evader motion planning, thereby significantly improving maneuvering efficiency. Notably, the evader must simultaneously accomplish evasion and terminal mission objectives, imposing more stringent energy management constraints than the pursuer. Consequently, accurate prediction of the pursuer's threat area is crucial for the evader's maneuver planning.

Flight vehicle motion prediction problems are typically cast as trajectory prediction tasks. Early research focused primarily on fixed ballistic trajectory prediction [1]. However, advances in flight vehicle technology have equipped them with increasingly sophisticated maneuvering capabilities, introducing substantial nonlinearities and uncertainties into trajectory prediction. This has motivated the development of novel trajectory prediction methodologies. A model-based strategy entails analyzing flight vehicle dynamics and subsequently predicting parameters of the resulting model. [2] accomplished parameter prediction via the Hough transform, while [3] employed empirical mode decomposition of aerodynamic acceleration.

Although prediction methods based on flight vehicle dynamics models can provide relatively stable prediction accuracy, their performance heavily relies on the fidelity of the underlying dynamical model. In adversarial flight scenarios, however, vehicles operate as non-cooperative targets with time-varying dynamics that are difficult to identify accurately. Consequently, data-driven approaches requiring no prior system knowledge have been increasingly adopted. [4] proposed a pipeline-based prediction model employing Long Short-Term Memory (LSTM) networks for parameter estimation. To address multiple flight modes, [5] utilized Recurrent Neural Networks for mode classification and subsequently performed trajectory prediction via LSTM. [6] also applied phase space reconstruction with radial basis function neural networks for trajectory prediction. Building on these developments, several studies have extended their scope to threat assessment. [7] developed an intruder trajectory predictor using sliding-window polynomial fitting to detect potential collisions through a static protected-zone model. [8] employed a Conditional Variational Autoencoder to forecast space object trajectories for subsequent threat evaluation.

However, the aforementioned methods do not explicitly account for environmental uncertainty. Consequently, probabilistic modeling approaches have received increasing attention. [9] incorporated Gaussian mixture models to enhance robustness under complex uncertainties, though their approach still required prior dynamical knowledge to extract the Koopman operator. As a data-driven probabilistic model, the Gaussian process (GP) offers strong noise suppression without prior model knowledge, prompting its application to flight vehicle trajectory prediction. The GP model has been employed to compensate for prediction errors in conventional methods such as dynamical modeling or Kalman filters [10, 11], or to predict model disturbances directly [12], thereby improving accuracy. Alternatively, the GP can directly predict flight vehicle trajectories [13].

The application of the GP has significantly enhanced trajectory prediction robustness in high-noise environments. However, existing studies [10–13] assume homoscedastic noise. In pursuer-evader games, noise levels typically vary with the relative distance between adversaries. Modeling this as constant fails to capture true noise characteristics, degrading prediction accuracy. Furthermore, existing threat area prediction methods typically adopt a two-stage approach: trajectory prediction followed by threat detection, which relies heavily on the accuracy of underlying trajectory estimates. To address these limitations, this paper proposes a pursuer threat area prediction algorithm based on heteroscedastic sparse variational Gaussian process (HSVGP). The main contributions are as follows:

(1) To address state-dependent noise in the pursuit-evasion game, we propose an HSVGP-based prediction model.

By embedding a Radial Basis Function (RBF) network into the likelihood function, the model captures heteroscedastic noise characteristics, thereby enhancing prediction accuracy in complex noise environments.

(2) Building upon trajectory prediction, we incorporate given maneuver plans to estimate the closest point of approach (CPA) between adversaries, directly leveraging the model's uncertainty quantification capabilities to provide probabilistic threat area (TA) predictions.

The remainder of this paper is organized as follows. Section 2 formulates the pursuit-evasion scenario. Section 3 details the proposed HSVGP-based threat area prediction algorithm. Section 4 presents simulation results and analysis. Finally, Section 5 concludes the paper.

2 Pursuit-Evasion Scenario

The pursuit-evasion scenario involves an evader $\mathcal{E}$ and a pursuer $\mathcal{P}$, as illustrated in Fig. 1. The relative distance is denoted by r, with initial separation r_0. The velocity vectors of $\mathcal{E}$ and $\mathcal{P}$ are $\boldsymbol{v}_{\mathcal{E}}$ and $\boldsymbol{v}_{\mathcal{P}}$, respectively. Acceleration is constrained to directions perpendicular to the respective velocity vectors, thereby altering the heading while maintaining constant speed. O denotes the inertial reference frame. $O_{\mathcal{P}_4}$ and $O_{\mathcal{E}_4}$ represent the line-of-sight (LOS) coordinate frames fixed to $\mathcal{P}$ and $\mathcal{E}$, respectively. The corresponding LOS angles are $q_{\mathcal{P},\epsilon}$, $q_{\mathcal{P},\beta}$ for $\mathcal{P}$ and $q_{\mathcal{E},\epsilon}$, $q_{\mathcal{E},\beta}$ for $\mathcal{E}$.

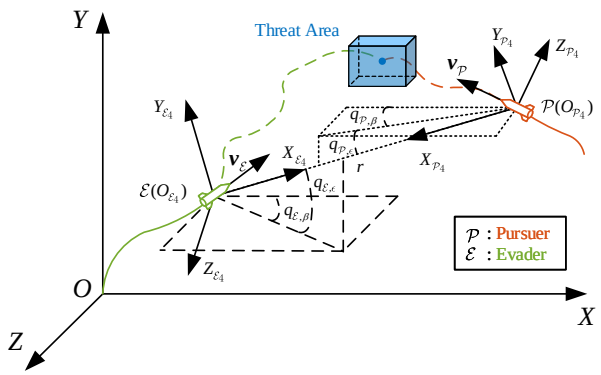

Fig. 1: Pursuit-evasion scenario

To evaluate the threat area prediction performance of $\mathcal{P}$, $\mathcal{E}$ adopts a random maneuvering strategy with lateral acceleration given by

$$a_{\mathcal{E}_y}, a_{\mathcal{E}_z} \sim \mathcal{U}(-a_{\mathcal{E},\max}, a_{\mathcal{E},\max}), \tag{1}$$

where $a_{\mathcal{E},\max}$ denotes the maximum lateral acceleration of $\mathcal{E}$. $\mathcal{P}$ employs the proportional navigation guidance (PNG) law given by Eq. (2) to pursue $\mathcal{E}$.

$$\boldsymbol{a}_P = \left[0, N_\epsilon \left| \dot{\boldsymbol{d}}_{\mathcal{P}} \right| \dot{q}_{\mathcal{P},\epsilon}, -N_\beta \left| \dot{\boldsymbol{d}}_{\mathcal{P}} \right| \dot{q}_{\mathcal{P},\beta}\right]^\top, \tag{2}$$

where $\boldsymbol{d}_{\mathcal{P}}$ denotes the relative position vector of $\mathcal{E}$ with respect to $\mathcal{P}$, and N_ϵ and N_β are the proportional navigation coefficients.

The onboard sensors of $\mathcal{E}$ measure the relative distance and LOS angles to $\mathcal{P}$, namely r, $q_{\mathcal{E},\epsilon}$, and $q_{\mathcal{E},\beta}$. The measurement model is given by

$$\begin{cases} r = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2} + \nu_r, \\ q_{\mathcal{E},\beta} = -\arctan\left(\dfrac{\Delta z}{\Delta x}\right) + \nu_a, \\ q_{\mathcal{E},\epsilon} = \arctan\left(\dfrac{\Delta y}{\sqrt{\Delta x^2 + \Delta z^2}}\right) + \nu_a, \end{cases} \tag{3}$$

where $\Delta x = p_{\mathcal{P}_x} - p_{\mathcal{E}_x}$, $\Delta y = p_{\mathcal{P}_y} - p_{\mathcal{E}_y}$, and $\Delta z = p_{\mathcal{P}_z} - p_{\mathcal{E}_z}$ represent the relative position components between $\mathcal{P}$ and $\mathcal{E}$. $\nu_r \sim \mathcal{N}(0, \sigma_r^2(r))$ and $\nu_a \sim \mathcal{N}(0, \sigma_a^2(r))$ denote distance and angle measurement noise, respectively, with standard deviations dependent on r. In this scenario, the noise standard deviations are assumed to be

$$\begin{cases} \sigma_r(r) = \min\left(\dfrac{r}{r_0}, 0.4\right)\sigma_{r,\max}, \\ \sigma_a(r) = \min\left(\dfrac{r}{r_0}, 0.4\right)\sigma_{a,\max}, \end{cases} \tag{4}$$

where $\sigma_{r,\max}$ and $\sigma_{a,\max}$ denote the maximum standard deviations of ν_r and ν_a, respectively.

Based on $\mathcal{E}$'s measurements of r, $q_{\mathcal{E},\epsilon}$, and $q_{\mathcal{E},\beta}$, along with its own position, the measured position of $\mathcal{P}$ is computed as

$$\begin{cases} p_{\mathcal{P}_x} = r\cos(q_{\mathcal{E},\epsilon})\cos(q_{\mathcal{E},\beta}) + p_{\mathcal{E}_x}, \\ p_{\mathcal{P}_y} = r\sin(q_{\mathcal{E},\epsilon})\sin(q_{\mathcal{E},\beta}) + p_{\mathcal{E}_y}, \\ p_{\mathcal{P}_z} = -r\cos(q_{\mathcal{E},\epsilon}) + p_{\mathcal{E}_z}. \end{cases} \tag{5}$$

3 Threat Area Prediction for the Pursuer Using HSVGP

The proposed HSVGP-based threat area prediction method addresses distance-dependent heteroscedastic noise in pursuit-evasion scenarios. An RBF network estimates noise levels and is embedded within the Sparse Variational Gaussian Process (SVGP) framework. By incorporating the evader's given maneuver strategy, HSVGP predicts the CPA and probabilistic TA. The overall framework is illustrated in Fig. 2.

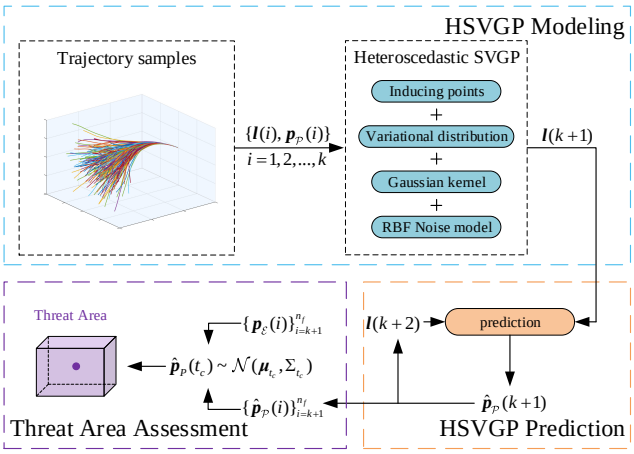

Fig. 2: Framework of threat area prediction using HSVGP

3.1 HSVGP with RBF Network

GP is a nonparametric modeling method that assumes the model output $f(\boldsymbol{x})$ follows a Gaussian process

$\mathcal{GP}(0, k(\boldsymbol{x}, \boldsymbol{x}'))$, where the input and the covariance function of the Gaussian process are denoted by $\boldsymbol{x} \in \mathbb{R}^d$ and $k(\boldsymbol{x}, \boldsymbol{x}')$, respectively. The true model output is generally unavailable, and only noisy observations can be obtained. The observation model is given by

$$y = f(\boldsymbol{x}) + \nu_f, \tag{6}$$

where $\nu_f \sim \mathcal{N}(0, \sigma_f^2)$ is Gaussian noise.

As a supervised learning method, the Gaussian process has a training dataset consisting of input data $\boldsymbol{x}_i \in \mathbb{R}^d$ and observations $y_i \in \mathbb{R}$, denoted as $\mathcal{D} = \{\langle \boldsymbol{x}_i, y_i \rangle\}_{i=1}^n$. The model complexity grows cubically with the number of data points, which leads to prohibitive computational cost when the dataset is large. To address this issue, variational inference [14] was introduced into the Gaussian process framework, resulting in the SVGP method [15].

SVGP significantly reduces model computational complexity by introducing a set of inducing points $X_m \in \mathbb{R}^{m \times d}$ with $m \ll n$, and by using a variational distribution $q(\boldsymbol{u}) \sim \mathcal{N}(\boldsymbol{\mu}_m, \Sigma_m)$ to approximate the true posterior over the inducing variables. The quality of the variational approximation is measured by the KL divergence between the variational distribution $q(\boldsymbol{u})$ and the true posterior $p(\boldsymbol{u} \mid \boldsymbol{y})$, as shown in Eq. (7). A smaller KL divergence indicates a better approximation of the posterior informed by the observed data.

$$\begin{aligned} \mathrm{KL}(q(\boldsymbol{u}) \parallel p(\boldsymbol{u} \mid \boldsymbol{y})) = \mathrm{KL}(q(\boldsymbol{u}) \parallel p(\boldsymbol{u})) + \log p(\boldsymbol{y}) \\ - \mathbb{E}_{q(\boldsymbol{u})}[\log p(\boldsymbol{y} \mid \boldsymbol{u})]. \end{aligned} \tag{7}$$

However, Eq. (7) is intractable to compute directly [14]. Therefore, it is reformulated to obtain the evidence lower bound (ELBO) as

$$\mathrm{ELBO} = \mathbb{E}_{q(\boldsymbol{u})}[\log p(\boldsymbol{y} \mid \boldsymbol{u})] - \mathrm{KL}(q(\boldsymbol{u}) \parallel p(\boldsymbol{u})). \tag{8}$$

Thus, the problem of minimizing Eq. (7) is transformed into maximizing the ELBO. Within the SVGP framework, the predictive distribution at a test point $\boldsymbol{x}_*$ is obtained by marginalizing over the variational posterior $q(\boldsymbol{u})$, which is learned from the training data $(X, \boldsymbol{y})$:

$$p(f_* | \boldsymbol{x}_*, X, \boldsymbol{y}) \approx \int p(f_* | \boldsymbol{x}_*, \boldsymbol{u}) q(\boldsymbol{u}) d\boldsymbol{u}, \tag{9}$$

where $X = \{\boldsymbol{x}_i\}_{i=1}^n$ denotes the training inputs. Evaluating Eq. (9) yields the mean and variance as

$$\mu(\boldsymbol{x}_*) = \boldsymbol{k}_{*X_m} K_{X_m X_m}^{-1} \boldsymbol{\mu}_m, \tag{10}$$

$$\begin{aligned} \sigma^2(\boldsymbol{x}_*) = & k_{**} - \boldsymbol{k}_{*X_m} K_{X_m X_m}^{-1} \boldsymbol{k}_{X_m *} \\ & + \boldsymbol{k}_{*X_m} K_{X_m X_m}^{-1} \Sigma_m K_{X_m X_m}^{-1} \boldsymbol{k}_{X_m *}. \end{aligned} \tag{11}$$

In the pursuit-evasion game, the evader uses its sensors to obtain the distance and angle measurements of the pursuer and then computes the pursuer position using Eq. (5). The measurement noise is dependent on the input space, exhibiting heteroscedastic characteristics. However, the standard GP treats the noise as a constant independent of the input space, which cannot accurately capture the true measurement noise behavior. To account for the correlation between noise and the input space, a heteroscedastic observation model is reformulated as

$$y = f(\boldsymbol{x}) + \nu_f(\boldsymbol{x}), \quad \nu_f(\boldsymbol{x}) \sim \mathcal{N}(0, \sigma_f^2(\boldsymbol{x})), \tag{12}$$

where the standard deviation $\sigma_f(\boldsymbol{x})$ of the noise exhibits a nonlinear mapping relationship with the input $\boldsymbol{x}$. Considering that, in the pursuit-evasion game, the noise does not change abruptly and instead possesses relatively smooth characteristics, we parameterize the standard deviation of the noise model using an RBF network as

$$\sigma_f(\boldsymbol{x}) = \mathrm{Softplus}(\boldsymbol{\omega}^\top \boldsymbol{\phi}(\boldsymbol{x}) + b), \tag{13}$$

where $\mathrm{Softplus}(z) = \log(1 + e^z)$ ensures that the output standard deviation remains positive. The RBF feature vector is defined as $\boldsymbol{\phi}(\boldsymbol{x}) = [\phi_1(\boldsymbol{x}), ..., \phi_J(\boldsymbol{x})]^\top$ with $\phi_j(\boldsymbol{x}) = \exp\left(-\frac{1}{2}\left(\frac{\|\boldsymbol{x} - \boldsymbol{c}_j\|_2}{l_j}\right)^2\right)$ for $j = 1, ..., J$, where J denotes the number of RBF units, $\boldsymbol{c}_j$ represents the center of the j-th RBF unit, l_j is the associated length scale, and $\boldsymbol{\omega} \in \mathbb{R}^J, b \in \mathbb{R}$ are the weight and bias parameters of the linear output layer, respectively.

Since the noise is now input-dependent rather than constant, the conditional likelihood for each observation becomes

$$p(y_i \mid f(\boldsymbol{x}_i), \boldsymbol{x}_i) = \mathcal{N}(y_i \mid f(\boldsymbol{x}_i), \sigma_f^2(\boldsymbol{x}_i)). \tag{14}$$

Substituting this into the expected log-likelihood term in the ELBO yields

$$\begin{aligned} & \mathbb{E}_{q(\boldsymbol{u})}[\log p(\boldsymbol{y} \mid \boldsymbol{u})] \\ & \quad = \sum_{i=1}^n \mathbb{E}_{q(f(\boldsymbol{x}_i))} \left[\log \mathcal{N}(y_i \mid f(\boldsymbol{x}_i), \sigma_f^2(\boldsymbol{x}_i))\right], \end{aligned} \tag{15}$$

where $q(f(\boldsymbol{x}_i))$ is the marginal variational distribution induced by $q(\boldsymbol{u})$ through the GP prior. Thus, Eq. (8) becomes

$$\begin{aligned} \mathrm{ELBO} = & \sum_{i=1}^n \mathbb{E}_{q(f(\boldsymbol{x}_i))}[\log \mathcal{N}(y_i \mid f(\boldsymbol{x}_i), \sigma_f^2(\boldsymbol{x}_i))] \\ & - \mathrm{KL}(q(\boldsymbol{u}) \parallel p(\boldsymbol{u})). \end{aligned} \tag{16}$$

It is worth noting that after introducing the heteroscedastic noise model $\sigma_f(\boldsymbol{x})$, if both $\sigma_f(\boldsymbol{x})$ and $q(\boldsymbol{u})$ are left unconstrained, insufficient fitting of the noise network $\sigma_f(\boldsymbol{x})$ may cause the optimizer to attribute residual errors to increased functional uncertainty. This leads to an inflation of Σ_m, resulting in a systematic overestimation of predictive uncertainty. To mitigate this issue, we introduce a covariance trace regularization term into the ELBO, defined as

$$\mathcal{L}_{\text{reg}} = \lambda \cdot \mathrm{Tr}(\Sigma_m), \tag{17}$$

where λ is the regularization coefficient. The ELBO then becomes the Regularized Variational Objective (RVO) as

$$\begin{aligned} \mathcal{L}_{\text{RVO}} = & \sum_{i=1}^n \mathbb{E}_{q(f(\boldsymbol{x}_i))}[\log \mathcal{N}(y_i \mid f(\boldsymbol{x}_i), \sigma_f^2(\boldsymbol{x}_i))] \\ & - \mathrm{KL}(q(\boldsymbol{u}) \parallel p(\boldsymbol{u})) - \lambda \cdot \mathrm{Tr}(\Sigma_m). \end{aligned} \tag{18}$$

During model parameter optimization, except for the RBF centers $\boldsymbol{c}_j$, which are initialized using the training dataset $\mathcal{D}$ via K-means clustering, all other parameters including the noise network parameters $\boldsymbol{\omega}$, b, RBF length scales l_j, kernel hyperparameters, inducing points, and variational distribution parameters are jointly updated by maximizing $\mathcal{L}_{\text{RVO}}$.

3.2 Threat Area Prediction Based on HSVGP

The pursuer's target is the evader, and its motion state depends on the relative motion between the pursuer and the evader. This relationship is nonlinear [16], so a nonlinear model for pursuer trajectory prediction can be formulated as

$$\boldsymbol{p}_{\mathcal{P}}(k) = f(\boldsymbol{l}(k)) + \nu_f(\boldsymbol{l}(k)), \tag{19}$$

where $f(\cdot)$ represents the nonlinear functional relationship between the model input $\boldsymbol{l}(k)$ and the pursuer's position $\boldsymbol{p}_{\mathcal{P}}(k)$ at time step k, and $\nu_f(\boldsymbol{l}(k))$ denotes the noise associated with input $\boldsymbol{l}(k)$. As a sequential model for pursuer trajectory prediction, relevant state information that may influence the pursuer's position is taken into account, and $\boldsymbol{l}(k)$ is constructed as

$$\begin{aligned}\boldsymbol{l}(k) = [&\boldsymbol{p}_{\mathcal{P}}(k-1), \ldots, \boldsymbol{p}_{\mathcal{P}}(k-h), \boldsymbol{p}_{\mathcal{E}}(k), \ldots, \\ &\boldsymbol{p}_{\mathcal{E}}(k-g+1)]^\top,\end{aligned} \tag{20}$$

where $\boldsymbol{p}_{\mathcal{P}}(i), i = k-h, ..., k-1$ and $\boldsymbol{p}_{\mathcal{E}}(j), j = k-g+1, ..., k$ are the position time series of the pursuer and the evader, respectively. h and g denote the time window widths for the pursuer's and evader's position sequences, respectively. The evader's position sequence includes $\boldsymbol{p}_{\mathcal{E}}(k)$ at the same time step as the predicted $\hat{\boldsymbol{p}}_{\mathcal{P}}(k)$, because the pursuer's motion between time steps $k-1$ and k is instantaneously coupled with the evader's motion at the corresponding time. Since $\boldsymbol{p}_{\mathcal{E}}(k)$ at time step k is unavailable at time step $k-1$, it must be computed based on the evader's maneuvering strategy.

In practice, to reduce model complexity, the pursuer's position $\hat{\boldsymbol{p}}_{\mathcal{P}}(k)$ is not directly predicted. Instead, the difference $\Delta\hat{\boldsymbol{p}}_{\mathcal{P}}(k) = \hat{\boldsymbol{p}}_{\mathcal{P}}(k) - \boldsymbol{p}_{\mathcal{P}}(k-1)$ is predicted to indirectly obtain $\hat{\boldsymbol{p}}_{\mathcal{P}}(k)$. The pursuer trajectory prediction model then becomes

$$\begin{cases}\Delta\hat{\boldsymbol{p}}_{\mathcal{P}}(k) = f(\boldsymbol{l}(k)) + \nu_f(\boldsymbol{l}(k)), \\ \hat{\boldsymbol{p}}_{\mathcal{P}}(k) = \boldsymbol{p}_{\mathcal{P}}(k-1) + \Delta\hat{\boldsymbol{p}}_{\mathcal{P}}(k).\end{cases} \tag{21}$$

The pursuer trajectory prediction model described in Eq. (21) is then constructed using the HSVGP approach proposed in Section 3.1, with a Gaussian kernel selected as the model's covariance function, given by

$$k(\boldsymbol{x}_1, \boldsymbol{x}_2) = \theta_s \exp\big(-\frac{1}{2}(\boldsymbol{x}_1 - \boldsymbol{x}_2)^\top \Theta^{-1}(\boldsymbol{x}_1 - \boldsymbol{x}_2)\big), \tag{22}$$

where θ_s and $\Theta = \mathrm{diag}\left(\theta_1^2, \ldots, \theta_d^2\right)$ are the signal variance and length scales of the kernel function, respectively.

Assuming prediction starts at time step k, the HSVGP-based prediction model and Eq. (21) can be used to obtain $\hat{\boldsymbol{p}}_{\mathcal{P}}(k+1)$, based on trajectory data of both the pursuer and the evader up to and including time step k. On this basis, the model input required to predict the pursuer's position at time step $k+2$ can be constructed as

$$\begin{aligned}\boldsymbol{l}(k+2) = [&\hat{\boldsymbol{p}}_{\mathcal{P}}(k+1), \ldots, \boldsymbol{p}_{\mathcal{P}}(k-h+2), \\ &\boldsymbol{p}_{\mathcal{E}}(k+2), \ldots, \boldsymbol{p}_{\mathcal{E}}(k-g+3)]^\top,\end{aligned} \tag{23}$$

Subsequently, $\hat{\boldsymbol{p}}_{\mathcal{P}}(k+2)$ is obtained, where all $\boldsymbol{p}_{\mathcal{E}}$ after time step k are computed based on the evader's given maneuver strategy. Repeating this process iteratively yields the predicted trajectory $\hat{\boldsymbol{p}}_{\mathcal{P}} = \{\hat{\boldsymbol{p}}_{\mathcal{P}}(i)\}_{i=k+1}^{n_f}$. In scenarios where the evader and pursuer are moving toward each other, the prediction terminates at the time step when their separation distance transitions from decreasing to increasing. This time step is defined as the prediction end time step n_f. Meanwhile, using the evader's given maneuvers starting from time step k, an expected trajectory $\boldsymbol{p}_{\mathcal{E}} = \{\boldsymbol{p}_{\mathcal{E}}(i)\}_{i=k+1}^{n_f}$ can also be generated. The trajectories of both the pursuer and the evader beyond time step k are then converted into two continuous trajectories $\hat{\boldsymbol{p}}_{\mathcal{P}}(t), \boldsymbol{p}_{\mathcal{E}}(t), t \in [t_{k+1}, t_f]$ using linear interpolation, where t_{k+1} and t_f represent the start and end times of the prediction horizon.

After obtaining $\hat{\boldsymbol{p}}_{\mathcal{P}}(t)$ and $\boldsymbol{p}_{\mathcal{E}}(t)$, our objective is to determine the time step when the evader and the pursuer are closest to each other, defined as the CPA time step, which is given by

$$t_c = \underset{t \in [t_{k+1}, t_f]}{\mathrm{argmin}} \|\hat{\boldsymbol{p}}_{\mathcal{P}}(t) - \boldsymbol{p}_{\mathcal{E}}(t)\|_2. \tag{24}$$

At this time step, the pursuer's predicted position follows $\hat{\boldsymbol{p}}_{\mathcal{P}}(t_c) \sim \mathcal{N}(\boldsymbol{\mu}_{t_c}, \Sigma_{t_c})$, which represents the location where the evader is expected to experience the highest threat under the current maneuver plan. The predicted mean $\boldsymbol{\mu}_{t_c}$ is taken as the center of the threat area $\mathcal{R}$, and the extent of $\mathcal{R}$ is determined by the predictive uncertainty Σ_{t_c}. Since HSVGP tends to overestimate predictive uncertainty, $\mathcal{R}$ is defined as the area within $\pm\sigma_{t_c}$ around $\boldsymbol{\mu}_{t_c}$. The resulting predicted threat area is then given by

$$\mathcal{R} = \{\boldsymbol{p} \mid |p_i - \mu_{t_c,i}| \leq \sigma_{t_c,i}, i = x, y, z\}, \tag{25}$$

where $\boldsymbol{\mu}_{t_c} = [\mu_{t_c,x}, \mu_{t_c,y}, \mu_{t_c,z}]^\top$ denotes the predicted center of the threat area, i.e., the CPA, and $\sigma_{t_c,i}, i = x, y, z$ represent the standard deviation of the corresponding predicted distribution.

4 Simulation Experiments and Analysis

In the simulation, the initial distance between the evader and the pursuer is $r_0 \approx 180$ km. It is assumed that the engagement has already entered the terminal guidance phase of the pursuer at the start of the scenario, and the velocity components of both the pursuer and the evader perpendicular to the LOS have already been aligned. The maneuver parameters, such as the initial speeds V_0, maximum accelerations $a_{\max}$, and maneuver periods T, are listed in Table 1.

Table 1: Flight vehicle maneuverability configuration

Flight Vehicle	V_0 (km/s)	$a_{\max}$ (m/s^2)	T (ms)
Evader	6	98	50
Pursuer	2.5	58.8	15

Before constructing the HSVGP-based threat area prediction model for the pursuer, the trajectory sample sequences are preprocessed. The input window length is set to $h = g = 3$, resulting in training samples that use the states from the past three time steps as model inputs. The maximum standard deviations of the distance and angle measurement noises are $\sigma_{r,\max} = 500$ m and $\sigma_{a,\max} = 0.5$ mrad, respectively. The training of the prediction model is conducted in two stages. First, the predictor is pretrained using a dataset $\mathcal{D}$

consisting of 500 recent trajectories, with a learning rate of $\alpha = 0.01$, $n_{\text{iter}} = 10$ hyperparameter optimization iterations, a mini-batch size of 512, and $m = 400$ inducing points. Then, during the online prediction stage, before each prediction, the predictor is incrementally updated using the latest trajectory data together with 1,000 randomly selected samples from $\mathcal{D}$. The incremental update uses a learning rate of $\alpha_+ = 0.001$ and the same number of iterations as n_{iter}. The number of centers in the RBF network within the HSVGP is set to $J = 48$. The root mean square error (RMSE), defined as $\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} \|\boldsymbol{p}_{\mathcal{P}}^{i} - \hat{\boldsymbol{p}}_{\mathcal{P}}^{i}\|_2}$, is employed as the metric to evaluate the discrepancy between predicted and true results.

The evader, following a given random maneuvering strategy, initiates predictions of the pursuer's trajectory using both HSVGP and SVGP when the separation between the evader and the pursuer reaches 170 km. Based on the predicted trajectories of both vehicles, the CPA and TA are obtained, as shown in Fig. 3. The prediction errors of the CPA by HSVGP and SVGP are 26.69 m and 63.04 m, respectively. HSVGP achieves higher prediction accuracy under complex noise conditions, although its prediction of the TA tends to be relatively conservative.

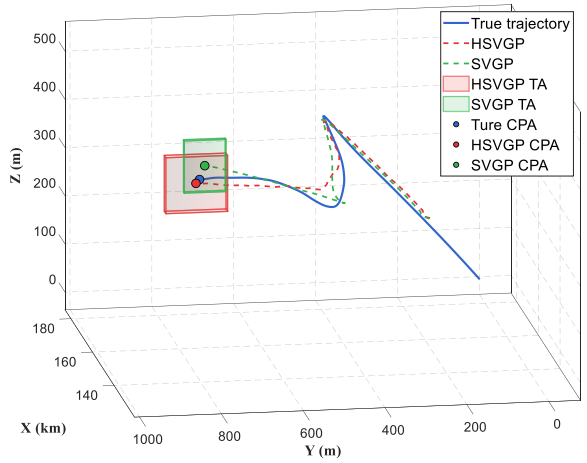

Fig. 3: Pursuer trajectory and threat area prediction results

To further validate the prediction capability of HSVGP under complex noise conditions, two sets of 1000-episode Monte Carlo simulations were conducted using both HSVGP and SVGP. In each episode, the model performs online incremental updates during the prediction process. These simulations include prediction results at distances r_{pred} of 170 km, 30 km, and 10 km. In the first simulation, the evader's initial position is randomly generated within a 500 m radius around the origin $(0, 0, 0)$, while its initial flight path angle and azimuth angle are fixed at $0°$. Since the pursuer's position is computed based on Eq. (5), the flight path angle and azimuth angle introduce nonlinear effects on positional noise. Therefore, varying these angles would increase the complexity of noise in trajectory prediction. In the second simulation, the evader's initial flight path angle and azimuth angle are uniformly randomized within the range of $[-2°, 2°]$, while the initial position setting remains the same as in the first simulation. The evolution of CPA prediction errors over the two Monte Carlo simulations is shown in Figs. 4-6. Table 2 presents the RMSE of the CPA prediction results computed every 200 episodes.

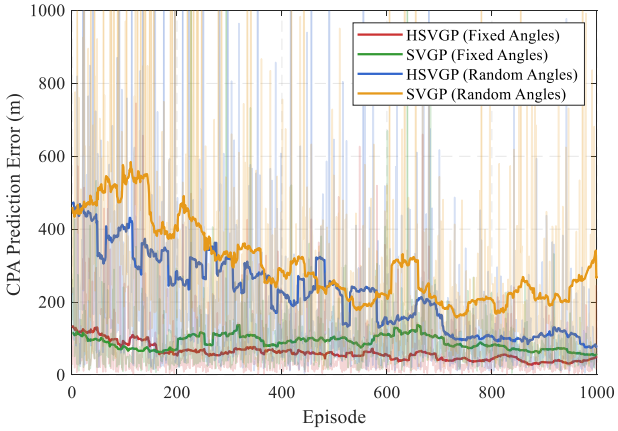

Fig. 4: X-axis CPA online prediction error variation in 1000-episode Monte Carlo simulations

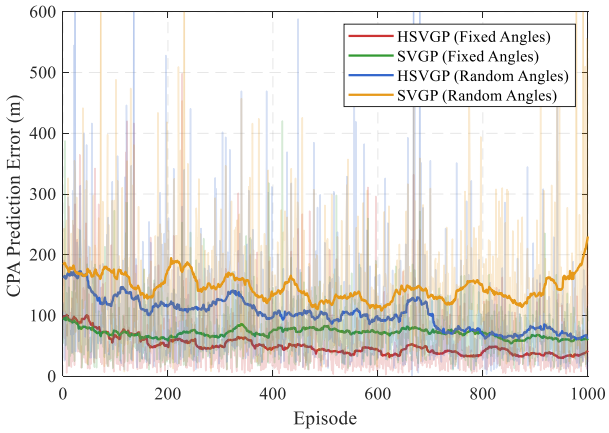

Fig. 5: Y-axis CPA online prediction error variation in 1000-episode Monte Carlo simulations

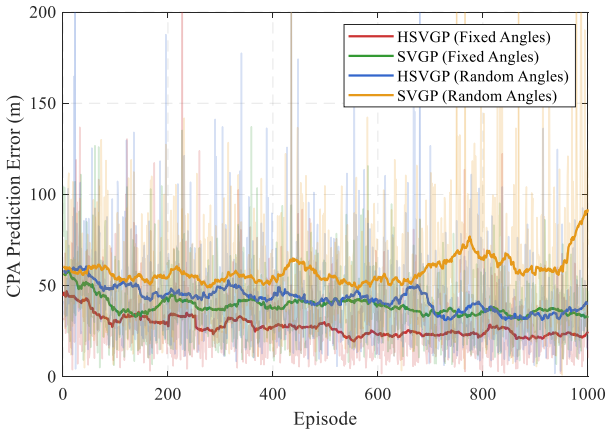

Fig. 6: Z-axis CPA online prediction error variation in 1000-episode Monte Carlo simulations

According to the results shown in Figs. 4-6 and Table 2, although both HSVGP and SVGP exhibit improved predic-

Table 2: CPA prediction error variation results at 170 km, 30 km, and 10 km under fixed-angle and random-angle scenarios

r_{pred} (km)	Episode	Fixed Angles		Random Angles	
		HSVGP	SVGP	HSVGP	SVGP
170	1-200	155.55	115.23	727.53	672.00
	201-400	102.75	203.80	629.25	500.18
	401-600	103.93	121.52	615.31	336.77
	601-800	79.41	153.79	222.22	419.53
	801-1000	61.43	79.76	152.47	344.46
30	1-200	106.20	90.03	172.69	189.29
	201-400	79.30	85.00	145.88	183.73
	401-600	67.06	90.90	130.03	152.44
	601-800	55.49	83.75	133.13	157.52
	801-1000	52.82	70.28	91.47	189.06
10	1-200	42.02	48.83	58.72	62.03
	201-400	41.09	44.28	52.61	58.79
	401-600	30.46	44.90	51.66	61.91
	601-800	27.33	39.08	48.22	68.50
	801-1000	28.45	38.17	40.44	78.11

tion accuracy through incremental training with new data, HSVGP clearly outperforms SVGP in complex noise environments. In the fixed-angle simulation, the performance gap between SVGP and HSVGP is relatively small. However, as the noise complexity increases in the random-angle simulation, the performance of SVGP, which is based on a homoscedastic noise assumption, degrades significantly, whereas HSVGP maintains consistently superior performance due to its enhanced capability to adapt to heterogeneous noise conditions.

5 Conclusion

The pursuit-evasion game constitutes a non-cooperative adversarial scenario where the evader's ability to predict pursuer maneuvers critically impacts mission effectiveness. This paper proposes a threat area prediction algorithm based on HSVGP. To address state-dependent measurement noise inherent in such scenarios, we design an input-dependent likelihood function using an RBF network to estimate noise levels from input states. Embedded within the SVGP framework, this enables joint optimization of noise and GP hyperparameters without requiring separate nonlinear noise modeling. Based on the predicted pursuer trajectory and the evader's planned maneuver, the CPA is first computed, and then the TA is derived as a probabilistic region centered at the CPA using the HSVGP's inherent uncertainty quantification.

Simulation results demonstrate that HSVGP exhibits superior adaptability to complex noise environments compared to standard SVGP, with performance advantages becoming more pronounced as noise heteroscedasticity increases. Future work will integrate the proposed method into an evasion decision-making framework, employing the predicted threat area as a core metric for strategy evaluation.

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